ECON100B - SECTION 17

IS Curve Pt. II

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ANNOUNCEMENTS/AGENDA

Announcements

- PSET 3 released: Due on Friday, 4/17

Agenda

- IS Curve Part II
- Problem Set 3

Question of the Day

Assume that my economy has a **real interest rate of 5%** and a **MPK of 4%.** If I wanted to **borrow 200 units** (dollars, for example) and invest it into capital, what would my expected return on capital be? What would my cost of borrowing be? What would my profit be? In this economy, would firms be incentivized to borrow? What would the impact on the short run output be in this economy?

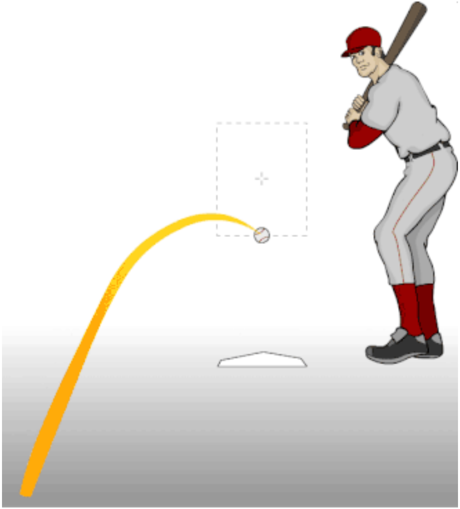
Question of the Day Answer

If I have a MPK of 4%, I will get a return of 8 units

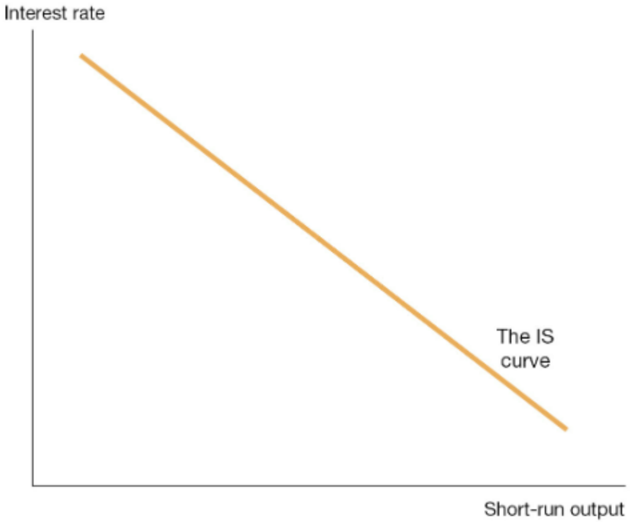
If I have a real interest rate of 5%, it will cost me 10 units to borrow

My profit would be -2 units, and in this case, most firms would not be incentivized to borrow, and there would be a negative impact on short run output in the economy

The IS Curve Part II



The IS Curve



Deriving the IS Curve

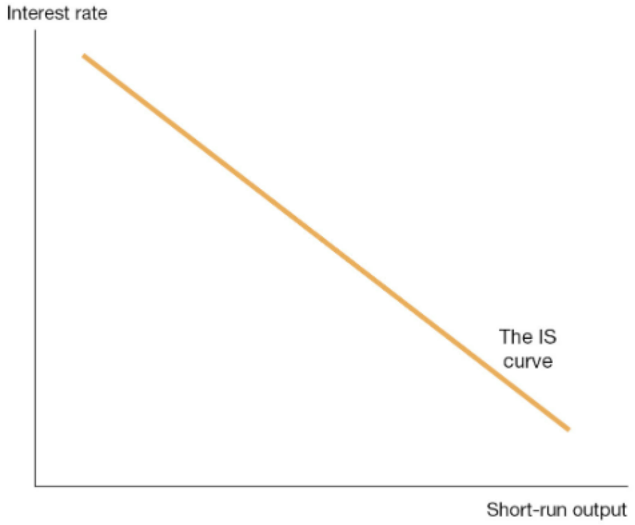
This gives us the following final equation

$$\tilde{Y}_t = \bar{a} - \bar{b}\left(R_t - \bar{r}\right),$$

Where:

$$\bar{a} \equiv \bar{a}_c + \bar{a}_i + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} - 1,$$

Moving along the IS Curve



IS curve – additional practice

1. Suppose that we experience a shock to $\overline{a_i}$ that increases the parameter by 2 percent, with all else held equal. What do we expect to happen to short run output? How would this be represented on the IS Curve?
2. Using the IS Curve, if we are told that $\bar{a}$ = 0, $\bar{b}$ = 0.4, R_t = 5%, $\bar{r}$ = 3%. What will out short run output be under this scenario? What is the interpretation of this scenario?
3. Say my $\bar{b}$ parameter increases to 0.6 from the previous scenario. What would short run output be under this new scenario? What does the higher "b" parameters represent?

IS curve – additional practice answers

1. We would expect short run output to increase by 2% as this is an aggregate demand shock. This would be represented by an outwards shift of the IS curve
2. Short run output of -0.8%. When our real interest rate is higher than the MPK, we expect a negative short run output due to lower investment
3. Short run output of -1.2%. This is because our sensitivity parameter has increased, i.e. our economy is more sensitive to changes in the real interest rate in relation to the MPK

Micro-foundations of the IS Curve

- We assume that there is “consumption smoothing” (see life-cycle model of consumption) across time, therefore, we say that consumption is relatively smooth relative to income
- People still want to have consistent consumption throughout time
- But this doesn’t always play out empirically: we sometimes see people increase/decrease their spending at times from shocks

- This leads into the idea of “multipliers” if aggregate consumption responds to temporary changes in income

The multiplier effect

Assume a multiplier between 0 and 1

$$\frac{C_t}{\bar{Y}_t} = \bar{a}_c + \bar{x}\tilde{Y}_t.$$

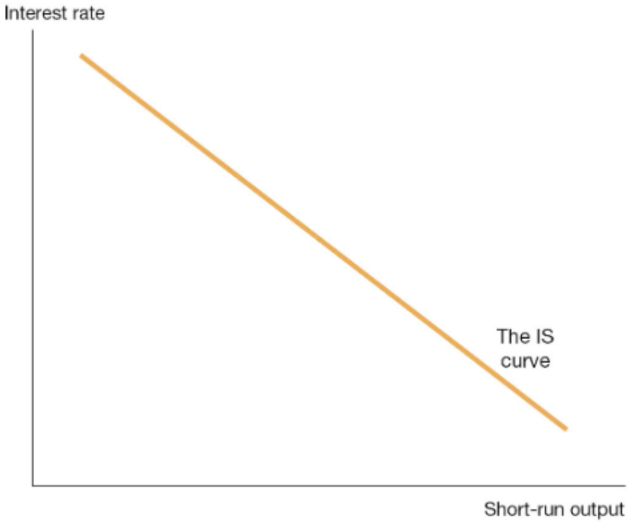
$$Y = \underbrace{\frac{1}{1-\bar{x}}}_{\text{multiplier}} \times \underbrace{\left[\bar{a} - \bar{b}\left(R_t - \bar{r}\right)\right]}_{\text{original IS curve}}.$$

What does this tell us?

- Any shock to aggregate demand or the interest rate terms has a larger effect than 1, therefore having a multiplier effect on short run output
- Short run output shocks have a multiplier effect by changing consumption behavior
- This leads to “vicious” or “virtuous” cycles

- The effect of other behaviors such as government spending are more ambiguous theoretically and empirically

IS Curve with the multiplier



Multiplier Practice

Using the IS Curve, if we are told that $\bar{a}$ = 0, $\bar{b}$ = 0.4, R_t = 5%, $\bar{r}$ = 3%. We are also told that our multiplier, or marginal propensity to consume is 0.2. What will out short run output be under this scenario? What is the interpretation of this scenario?

Multiplier Practice Answer

Our short run output under this scenario would be -1.0%, the multiplier effect finds that when we see a drop in short run output, this effect in compounded by changes to consumer behavior.

PSET 3, Q4

4. [12 points total] The IS curve, interest rates, and macroeconomic shocks.

Recall how the IS curve is a downward-sloping line in $\left(\tilde{Y}, R\right)$-space

$$\tilde{Y} = \bar{a} - \bar{b}\,(R - \bar{r}) \tag{4.1}$$

(a) [2 points] Graph the IS curve on the axes below, being careful to position it correctly so that you are *depicting no temporary shocks* to aggregate demand.

PSET 3, Q4

Suppose that $\bar{b} = 0.4$ and $\bar{r} = 0.03$ so that

$$\tilde{Y} = \bar{a} - 0.4\,(R - 0.03) \qquad (4.2)$$

In each of the following scenarios, discuss what happens to the IS curve, if anything, and to short-run output $\tilde{Y}$, if anything. Begin each scenario by assuming that there is no temporary shock to aggregate demand $\bar{a} = 0$ and that monetary policy is neutral, $R = \bar{r}$. Both may change.

(b) [2 points] Suppose consumers become spooked by falling stock prices and temporarily reduce consumption spending by 0.1 percentage point of potential GDP.

(c) [2 points] Suppose the Fed decides it must fight inflation (not shown here), and it raises the real interest rate to $R' = 0.04$.

PSET 3, Q4

(d) [2 points] Suppose Congress (the government) votes to go to war and passes a bill to raise government spending on military goods and services by \$0.3 trillion or roughly 1 percent of GDP, and the President signs the bill.

(e) [2 points] Suppose foreign economies temporarily increase their demand for U.S. exports by 0.3% of potential GDP.

(f) [2 points] Suppose the marginal product of capital $\bar{r}$ falls from $\bar{r} = 0.03$ to $\bar{r}' = 0.02$ but real interest rates remain at $R = 0.03$.